

AYAIR 2026 — THIRD EDITION

NeuroScope

Visual Deep Learning Builder — Learn by Building, Not by Reading



Team: DigiNeurons

Category: Education Enhancement

Nature: Group Project (8 Members)

Submitted: June 2026

Project Introduction

NeuroScope is an open-source visual deep learning builder that lets students and developers construct complete ML/DL models by dragging, dropping, and selecting components — no coding required. It transforms the way deep learning is taught by making every action educational: users build real, production-ready models while understanding every choice they make. NeuroScope addresses the critical gap between ML/DL theory and implementation across Africa, where the mentorship infrastructure that could bridge this gap simply does not exist.

Origin

NeuroScope was born from a real problem we encountered as trainees in the **Digilians Initiative** under Egypt's Ministry of Communications and Information Technology (MCIT). During our deep learning training in New Cairo, we watched colleagues struggle with the same challenge: they could write `model.fit()` and achieve accuracy scores, but could not explain what happened inside their models. The architecture was a black box. They copied architectures from GitHub without understanding why each layer existed. When training failed, they had no way to diagnose the problem.

This gap is systemic across Africa. The tools available — Netron (static 2D diagrams), TensorBoard (TensorFlow-locked), and TensorSpace.js (abandoned since 2019) — do not explain what layers do, detect architectural mistakes, or map visualization back to code. None of them teach you how to *build*. They show you the machine after it is built. They do not teach you how to build it. We decided to build the tool we needed ourselves.

Team — DigiNeurons

This is a **group project** submitted by the team leader on behalf of all eight members. The team is organized into three tracks: Data Analysis, Data Science, and Software Development.

#	NAME	ROLE	TRACK
1	Hazem Khaled	Team Leader · Data Analyst	Data Analysis
2	Yomna Ashraf	Data Analyst	Data Analysis
3	Yossef Shrif	Data Analyst	Data Analysis
4	Shahd Khairy	Frontend Developer	Software
5	Mohamed Wagdi	Backend Developer	Software
6	Ziad Mohamed	Backend Developer	Software
7	Yossef Safout	Backend Developer	Software
8	Mohamed Abdel Ghani	Data Scientist	Data Science

Problem Statement

Every year, millions of students across Africa begin learning machine learning and deep learning. They attend lectures, study theory, and watch tutorials. But when they sit down to build their first model, they face a wall. The gap between understanding what a convolutional layer is and actually writing one in code is enormous. Most students never cross it. They copy architectures from GitHub without understanding why each layer exists, run `model.fit()` without knowing what happens inside, and when training fails, they cannot diagnose the problem.

This is not a talent problem. It is an infrastructure problem. Across Africa, there are not enough senior ML engineers to guide every student. A student in rural Nigeria debugging a failing model at midnight has nowhere to turn. The mentorship gap means students are left alone with broken models and no way to understand why they broke.

Solution Overview

NeuroScope lets users construct complete deep learning models through a visual drag-and-drop interface. The model appears on screen as a **3D machine**: layers are connected blocks, extensions like optimizers and loss functions orbit the core engine with visible cables, and the entire architecture comes alive as the user builds it.

Every action teaches something. Click the optimizer extension and you see what Adam does differently from SGD. Add a convolutional layer and you understand how filters slide across

your input and generate feature maps. When a configuration might cause problems — such as using sigmoid in a deep network or a learning rate that is too high — NeuroScope flags it with a clear explanation and suggested fix.

The tool acts as an **automated mentor**, catching mistakes that a senior engineer would catch during code review — available anytime, anywhere, for free.

Key Features

Visual Drag-and-Drop Builder

Construct complete ML/DL models by selecting and connecting components — no coding required. The 3D visualization makes architecture tangible and intuitive.

Learn While Building

Every option includes plain-language explanations, recommended use cases, and consequences of each choice. Students learn by doing, not by reading documentation.

Production-Ready Code Export

Export clean Jupyter notebooks or Python files. Take to Google Colab, add your dataset, and run. The code follows best practices because every choice was guided.

Automated Architecture Review

Real-time detection of common mistakes and anti-patterns with severity ratings, explanations, and suggested fixes — like a senior engineer reviewing your code.

Develop Mode for Advanced Users

Inspect full model code, freeze or unfreeze individual layers, modify architectures at a granular level, and customize the head layer for specific classification tasks.

Multi-Model Support

Launches with CNN (deep learning). Future releases will support YOLO, ResNet, EfficientNet, and classical ML models — making NeuroScope a comprehensive builder for the entire ML/DL spectrum.

Technical Architecture

Product Platform

NeuroScope launches as a **web application** (primary product) with a companion **desktop application** that works both online and offline — ensuring access in any environment, from university labs to areas with no connectivity. A **VS Code extension** is planned for future release, allowing users to build and run models directly inside their development environment.

Technology Stack

Frontend

React 18 + TypeScript + Three.js (React Three Fiber) for 3D rendering + Zustand for state management + Tailwind CSS for styling

Backend

Python + FastAPI for API services + model parsers (ONNX, PyTorch, Keras, TensorFlow Lite) + analysis engine with 47+ rules

Infrastructure

Docker containerization + CI/CD pipeline + progressive web app (PWA) for offline capability

First Supported Model: CNN v16

A 16-layer convolutional neural network with **seven configurable extensions**:

EXTENSION	CATEGORY	OPTIONS
Optimizer	Training	SGD, Adam, AdamW, RMSprop
Activation	Training	ReLU, LeakyReLU, SiLU, Mish, GELU
Loss Function	Training	CrossEntropy, FocalLoss, LabelSmoothing
Learning Rate	Training	0.1, 0.01, 0.001, 0.0001
Batch Size	Functional	8, 16, 32, 64, 128
Epochs	Functional	50, 100, 200, 500
Augmentation	Data	None, Basic, Advanced, Custom

Each option includes educational content: what it is, when to use it, what happens if misconfigured, and the code it generates.

Impact & Alignment

5 LANGUAGES	0 COST	2.5B+ SPEAKERS REACHED
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NeuroScope is **free, open-source, and offline-capable**. It supports English, French, Arabic, Swahili, and Portuguese. It runs in any browser on any device, including low-end laptops and tablets. There is no cost, no subscription, no GPU requirement. Universities can deploy their own instances via Docker.

The tool aligns with **AU STISA 2034, Agenda 2063, and Egypt's MCIT digital transformation strategy**. It is inclusive because it works offline, in multiple languages, on any device. It is sustainable because it is open source and community-driven. It is efficient because it replaces months of trial-and-error learning with guided, interactive model building.

Sustainability

Released under the **MIT license**, NeuroScope is designed for community contribution. Analysis rules are YAML-configurable. Layer descriptions are structured data. Translations use JSON language files. New model architectures can be added through structured

definitions. The web app and desktop app form the foundation; a VS Code extension will bring the visual builder directly into the developer's coding environment.

"We built NeuroScope not as a competition entry, but as lasting infrastructure for deep learning education. The problems we faced as trainees are the same problems students face across Africa. NeuroScope does not just show you the machine. It teaches you how to build it."